TEAM 83 ***PRODUCT DESIGN CHALLENGE***



DESIGN REPORT

LOW PREP SOLINAS PRODUCT DESIGN CHALLENGE

All Models, Simulations, Codes, etc - *Drive link*

CONTENTS

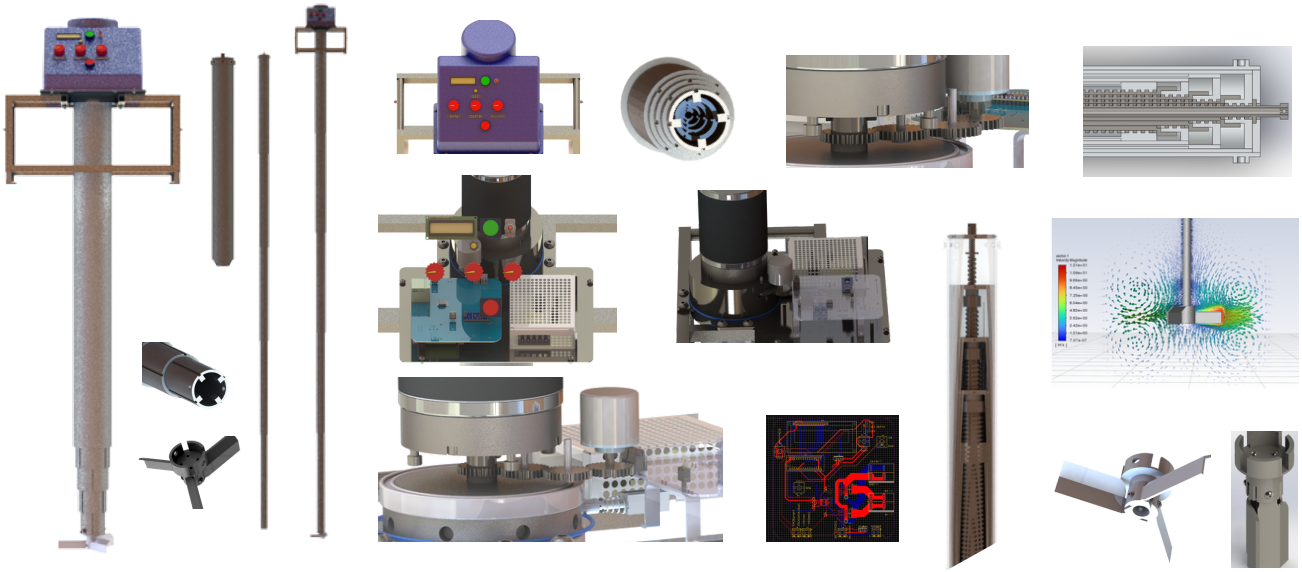
Title *Pg.no*

1. Problem Statement (Brief) 1
2. Our Solution 1
   - Product Description 2
   - Components 2
   - Extending Mechanism 2
   - Depth Capability 2
3. Flexibility and weight 2
   - Material & Weight 2
   - Centrifugal Retractable Blade 3
   - Blade changing Mechanism and advantages 3
   - Confirmation Via Ansys Simulation 3
4. Portability 3
   - Achievable Compactness 3
   - Self-Engaging Mechanism at Maximum Length 3
   - The mechanism for using it at any arbitrary length 4
   - Bill of Materials 4
5. User Friendly Interface 4
   - How to use 4
   - PCB Design and Electronics 4
   - MATLAB Simulink Representation 5
6. Water Proofing 5
   - Why Our Device is completely Waterproof ?: Key Factors 5
   - How to deal with Caviation 5
   - Other Additional Measures 5
7. Design Safety from failure 6
   - Determining Fatigue Factor of Safety by DE Goodman criteria 6
   - Confirmed Via FEA Simulation 6
   - Bearing Selection via SKF catalog 6
   - Why choose screw threaded mechanism over other alternatives ? 7
8. References 7

1. PROBLEM STATEMENT

The challenge is to design a compact and innovative shaft for an agitator mechanism that can homogenize solid material or waste with liquid substances in tanks with a depth of 5 meters. The shaft needs to be retractable to a length of 1 meter for portability and extendable to 5 meters to operate in a tank of 0.4 diameter. It should be user-friendly,portable and provide waterproofing and spark-proofing for safe operation in underwater or sludge storage tanks.

2. OUR SOLUTION

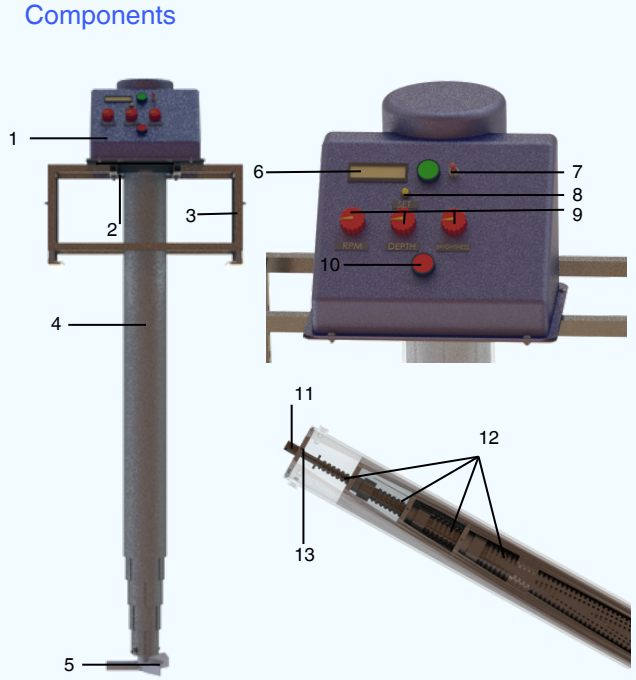


Product Description

A compact slurry mixing machine designed to be portable and easily operable. This can mix thick slurry from a depth of 1m to 5m using a mechanical extending mechanism. The user interface is simple and the electronics are well packed and accomodated for a safer usage even by inexperienced operators.

CAD File - *drive link* (Solidworks assembly)

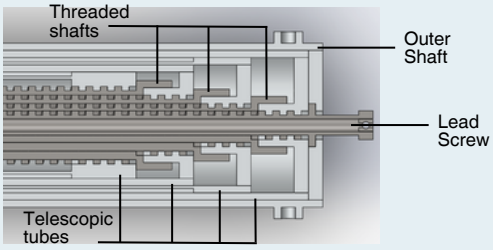
Components

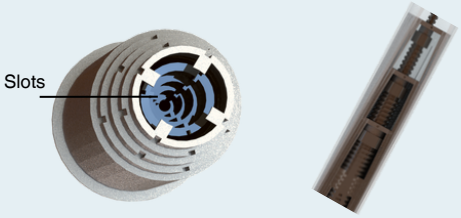


1 Central Console
2 central bearing
3 Stand
4 Main Shaft
5 Impellers
6 Display
7 Set rotation direction toggle
8 Depth Set Button
9 Rotary Knobs
10 Power OFF button
11 Motor coupler
12 Threaded Shafts
13 Motor shaft bearing
14 Rotary Knobs
15 Output to locking solenoid
16 Output to Drive motor
17 Motor driver IC
18 5v Voltage regulator IC
19 Microcontroller
20 SMPS
21 Grounded casing ofvSMPS
22 Motor with gearbox
23 Engaging mechanism
24 Rotary Encoder
25 Pulley
26 Backup Battery
27 PCB

Extending Mechanism

- The extending mechanism uses a lead screw and 3 threaded outer shafts which can slide on the lead screw and still transmit the rotation to the outer telescopic tubes which are arrested of their rotation mechanism.
- Extending: The lead screw rotates freely inside the outer shaft which is locked using a linear actuator.
- The threaded telescopic tubes rides the threads as the rotation of inner telescopic tubes are arrested by the slots with the outer tubes, but are free to move axially, thus imparting a linear motion.
- This motions occurs simulatneously in 4 tubes and thus the mechanism can extend from 1m to 5m in 100 rotations of the lead screw.
- If a desired depth is reached the linear actuator disconnects the outer shafts and lets the whole assembly rotate due to the self locking property of threads thus rotating the impeller attached to the inner most tube
- The impeller open and rotates only when the shafts is set at a desired depth.





Depth Capability

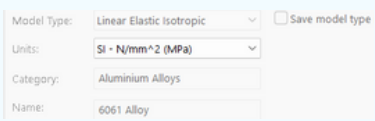
The shaft can extend from a length of 1m to 5m in **100 rotations** of the lead screw.
Time taken for full length : 50 seconds at 120 RPM

3. Flexibility and weight

The extending shaft is rigid and devoid of any unexpected flex. This mechanical locking and rotating mechanism makes it durable and simple at the same time. This mechanism is predictable and safe to use in deep tanks or holes. The manageable weight makes it easily portable

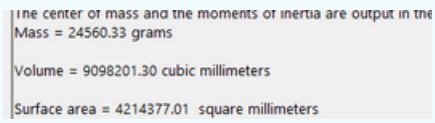
Material

Aluminium alloys have been used for strength and lowering weight



Weight

The total weight of the machine is around 25Kg, thus making it easily portable



Centrifugal Retractable Blade

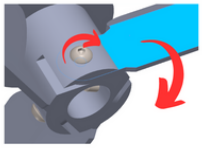
Typically, installing mixing impellers within sewage tanks presents challenges due to the highly viscous nature of slurry. Our solution involves a specifically designed mechanism that simplifies installation and removal, catering to the confines of small-diameter sewage tanks. Once submerged to the required depth and set in motion, the innovative blade design refracts efficiently, effectively mixing the slurry.
Based upon the literature study, we decided the length and thickness of the blade to be of 160mm and 3mm respectively. ***(Ref 1)***
Calculation of width of the blades -

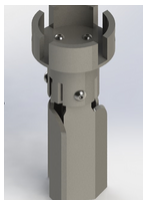
$$\dot{m} = 1181\left(\frac{d}{D}\right)^{-0.0226} \quad \& \quad \dot{m} = \frac{a\left(\frac{W}{D}\right)}{1+b\left(\frac{W}{D}\right)}$$

W = 38 mm or 0.038 m

m= mass flow rate
d= shaft diameter = 0.02 m
D= impeller diameter = 0.16 m
a= 20,000 kg/s
b=12
W= width of blade
(Ref 2)

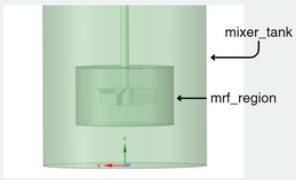

Fully Refracted condition


Folding direction

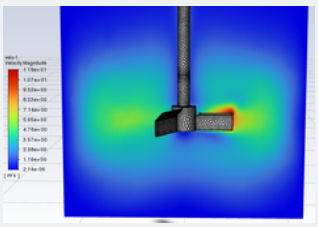

Folded conditon

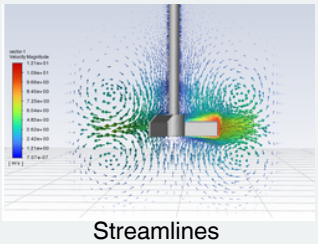
Confirmed Via Ansys CFD Simulation -

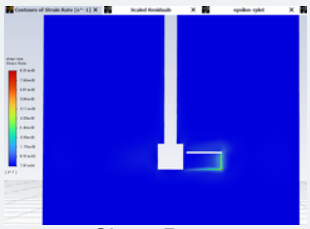
- For CFD, Moving Reference Frame(MRF) approach is followed. The shown mrf region is rototed around the body axis while the body remains still during the simulation phase.
- The reference frame follows the Navier- Stokes equation with coriolis and centrifugal forces.



Geometry for analysis


Velocity Contour


Streamlines

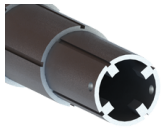

Shear Rate

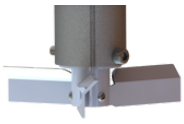
Conclusion

From CFD results, it is found that the impeller effectively mixes the slurry within a 250 mm vertical range. To optimize mixing efficiency the impeller can be operated at different heights.

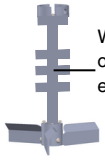
Blade changing Mechanism

Individual blades and the blade set can be changed easily as they are attached using easily accessible bolts. The coupler attaching to the telescopic tube allows for use of impellers of any design.






can easily replaced damage blades


We can any type of blade / blade extension now

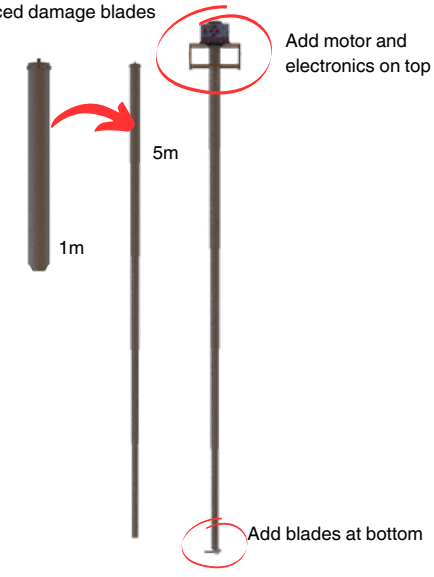
Blade changing Advantages

- Can have multiple different blades for various dimensions of slurry tanks
- No need to change the entire shaft if the blade gets damaged, thus saving costs tremendously
- Due to the centrifugal opening of blades, long blades can be used for a system which has narrow passage way but larger underground reservoir at the end

4. Portability

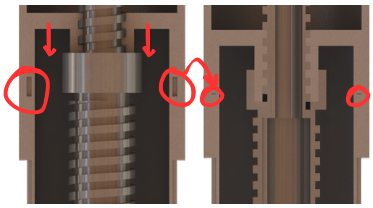
Achievable Compactness

The whole assembly including the motor and impellers can be packed into a box of 1.5m and 0.5m thus proving its compact size. Just weighing around 24 Kg makes it easy to use and shift between work locations.The extending shaft mechanism enables to pack an agitator mechanism capable of working at 5m into these compact dimensions.



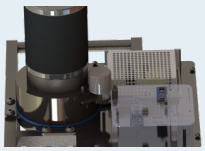
Self-Engaging Mechanism at Maximum Length

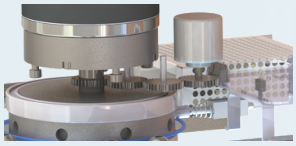
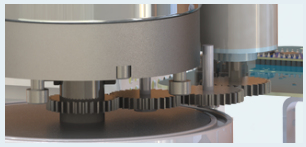
- The mechanism is designed such that the blades will only activate after reaching maximum length automatically so that the walls of the tank do not get damaged.
- The actuator makes the process of engaging the blade effortlessly and allows for proper control of the RPM of the motor to control the extending speed. The display shows the depth and the operator can engage the blades manually at any desired depth.


(At max length)

The engaging mechanism for using it at any arbitrary length

The rotary encoder reads the rotation of the shaft when extending thus calculating the length. When maximum depth is reached the blade engages automatically, while the operator can override to engage the blade at any arbitrary depth for proper working.



Bill of Materials

S.No.	Quantity	Item	Cost in Rupees	S.No.	Quantity	Item	Cost in Rupees
1	1	100P	2	14	1	CONN-TH_2P-P5.00_DB128L-	15
2	1	DISPLAY-LCD-16X2	100	15	5	Aluminium Telescopic Tubing	11000
3	8	10k	8	16	4	Threaded Tubing	3500
4	1	SS13D07VG2	10	17	1	Impeller coupling and blades	1200
5	1	SN74HC244DBR	50	18	1	Stand	800
6	2	BTN8982TAAUMA1	100	19	1	Base	1600
7	1	LM7805T	10	20	1	Motor	5000
8	1	ARDUINO_NANO	220	21	1	Gearbox	2000
9	3	DB128L-5.0-2P-GY-S	45	22	LS	Nut and Bolts	400
10	3	SW-TH_EC11G1560421	120	23	2	Bearings	1200
11	2	CONN-TH_2P-P10.00_JL271R-10002G01	60	24	1	Washers	250
12	1	SW-TH_1AD1T1B4M2QES	10	25		***Overhead***	12000
13	1	TO-220-3_L9.9-W4.5-P2.54-L	30				
		Total =	**39730**				

5. USER FRIENDLY INTERFACE

How to use

Operation

Step 1: Push slider from 'off' to 'set_depth' state.

Step 2: Set depth by adjusting the depth knob, (you can see the input(desired) depth and current depth attained on the screen)

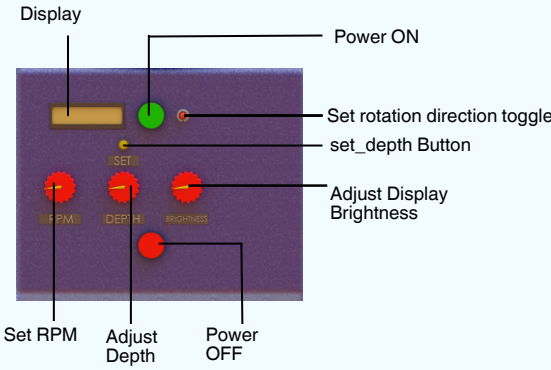
Step 3: Once desired depth is attained, push the slider from 'set_depth' state to 'on' state. Set direction of rotation using toggle switch.
(you can now see the depth and RPM on the screen).

Turning off the device

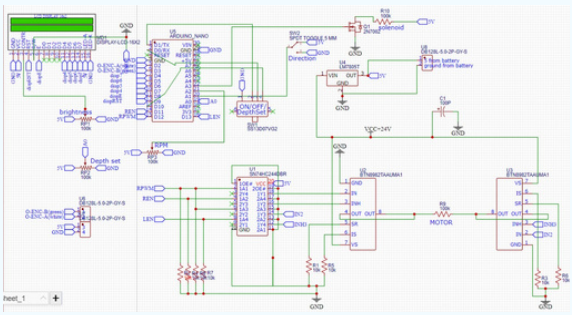
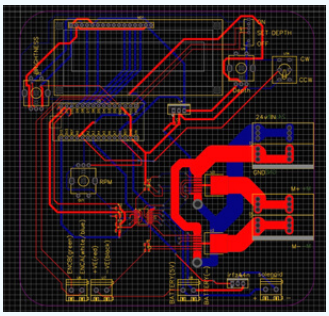
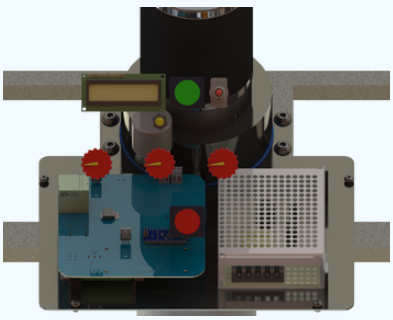
Step 1: Set RPM to 0.

Step 2: Push slider from 'on' to 'set_depth' state, and set depth to 0.

Step 3: Push slider from 'set_depth' to 'off' state.

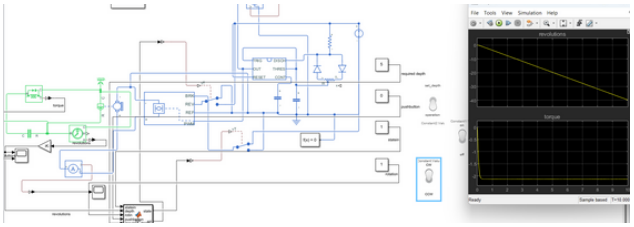


PCB Design and electronics

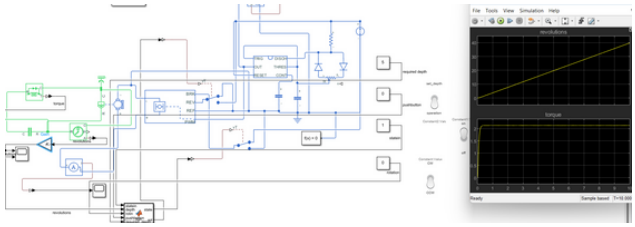




Note: that in the design, we have used 2 switches to select any of the 3 modes, whereas in the circuit and code we have used a single switch with 3 states. Any one of the following can be implemented int he final design, based preference

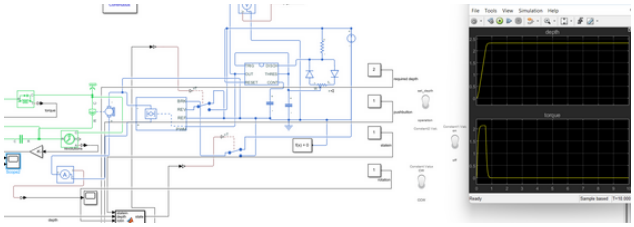
Simulink Represention



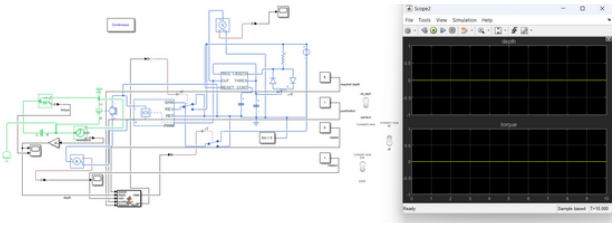
(1) operation mode (clockwise)



(2) operation mode (counter clockwise)



(3) depth set to 2m.



(4) off state

the Simulink diagram shows 4 different operational states.
(1) when on state in enabled and toggle is flipped to clockwise - we see a negative Rotation of the whole shaft, and torque for the whole time
(2) we can see the same thing but in the opposite direction
(3) depth has been given as 2m, so the shaft will extend up to 2m depth. note that whenever depth input is changed, the shaft will attain that depth only in the 'set_depth' mode, and will cu toff power to the motor upon reaching the desired depth.
(4) off state: cuts off power to the motors, and locks the shaft so that its length dosen't change.

MATLAB Simulink file - *drive link* PCB Design file - *drive link* Code - *drive link*

6. WATER PROOFING

Why Our Device is completely Waterproof ?: Key Factors

- The most effective approach to waterproofing is to eliminate the need for waterproofing altogether by devising a design that prevents direct exposure of electrical components to liquid-prone areas within machines.
- In the context of this mechanism, **the extending mechanism is entirely mechanical, ensuring that all electrical components are positioned** on the central operating console, which is located **outside the tank**.
- This failproof configuration guarantees complete water and spark-proof functionality as all electronics are outside
- Moreover, we have all **joints sealed** and can use Slip Rings and Sealed Bearings to prevent water from entering
- We have **Tight Tolerances** in the design as well

How to deal with Cavitation

- Our device uses **wear-resistant coating** such as Metaline to prevent cavitation
- The walls of the extending telescoping shaft is quite thick to prevent shaft damage in long run
- Risk is mainly in the blades which can easily be replaced with minimum cost. see **blade changing mechanism** (page 3)
- We can also use *bionic propellers* to reduce cavitation (Con- Tough geometry to manufacture)
- Use **anti-vibration** mounts to reduce mechanical stress on seals and joints.

Other Additional Measures

- Pre-testing and regular maintenance - Perform rigorous testing under various water exposure scenarios beforehand and while deployed check for regular care
- Smooth Surface Finish: Ensure a smooth surface finish on all components to minimize water retention.
- Ingress Protection (IP) rating Compliance: Aim for an IP rating that aligns with your device's usage environment. IP ratings specify the level of protection against solids and liquids, providing a standardized measure.
- Hydrophobic Materials: These materials naturally repel water, minimizing the chances of water adherence to surfaces.

7. DESIGN SAFETY FROM FAILURE

We have selected the innermost rotating threaded shaft (which is coupled to the motor) as a Critical Region for failure. When the shaft is in maximum extension i.e. 5m, then the shaft will be most likely to fail. Also we are taking case of Maximum Torque applied by the motor

So considering that, There will be an axial Force Fa (Due to mass of entire mechanism, "mg")=(24,78 kgX9.8)Newton and and maximum torque = 350 NM

Determining Fatigue Factor of Safety by DE Goodman criteria

(A) For the calculation of standard alternating and midrange stresses

$$\sigma_a = K_f \frac{M_a c}{I} \qquad \sigma_m = K_f \frac{M_m c}{I} \qquad \sigma_a = K_f \frac{32 M_a}{\pi d^3} \qquad \sigma_m = K_f \frac{32 M_m}{\pi d^3}$$

$$\tau_a = K_{fs} \frac{T_a c}{J} \qquad \tau_m = K_{fs} \frac{T_m c}{J} \qquad \tau_a = K_{fs} \frac{16 T_a}{\pi d^3} \qquad \tau_m = K_{fs} \frac{16 T_m}{\pi d^3}$$

(B) For calculating von-mises stress

$$\sigma'_a = (\sigma_a^2 + 3\tau_a^2)^{1/2} = \left[\left(\frac{32 K_f M_a}{\pi d^3}\right)^2 + 3\left(\frac{16 K_{fs} T_a}{\pi d^3}\right)^2\right]^{1/2}$$

$$\sigma'_m = (\sigma_m^2 + 3\tau_m^2)^{1/2} = \left[\left(\frac{32 K_f M_m}{\pi d^3}\right)^2 + 3\left(\frac{16 K_{fs} T_m}{\pi d^3}\right)^2\right]^{1/2}$$

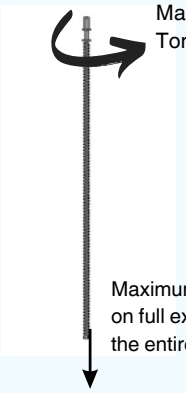
(C) Failure Criterian - Modified Goodman Theory

$$\frac{1}{n} = \frac{\sigma'_a}{S_e} + \frac{\sigma'_m}{S_{ut}}$$

$$\frac{1}{n} = \frac{16}{\pi d^3}\left\{\frac{1}{S_e}\left[4(K_f M_a)^2 + 3(K_{fs} T_a)^2\right]^{1/2} + \frac{1}{S_{ut}}\left[4(K_f M_m)^2 + 3(K_{fs} T_m)^2\right]^{1/2}\right\}$$

In or case for Al 6061
Sut =310 Mpa
Sy = 260 MPa
d = 20mm
J = 1.5708 cm^4
I = 0.7854 cm^4
Kt = 2.09
v = 0.33
q= notch sensitvity

$$K_f = 1 + q(K_t - 1) \quad \text{or} \quad K_{fs} = 1 + q_{shear}(K_{ts} - 1)$$

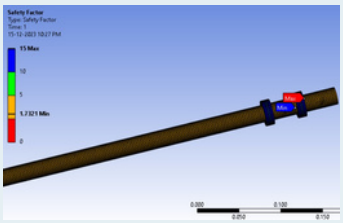


(Critical breaking region FBD)

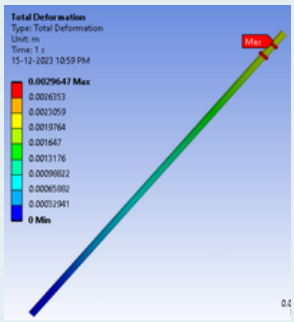
Factor of safety - 1.84 Theoretical

Confirmed via FEA simulation

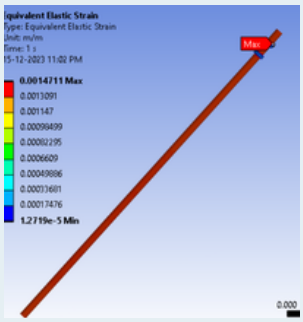
Material Selected - Al 6061, T6
Ultimate tensile strength, Sy = 260 MPa



Factor of safety - 1.7321 (at most critical point)



Total Deformation



Equivalent Elastic Strain

Bearing Selection via SKF catalogue

$$F_e = (XVF_r + YF_a)C_s \qquad \left(\frac{C}{F_e}\right)^a = L$$

Here Fa = 24.78 kg or Fa = 242.84 N Bore dia = internal dia of threaded shaft = 20mm

Here Fa = 24.78 kg or Fa = 242.84 N
Co = 5000, Cs =1 , C= 9950 , a=3 (deep groove ball bearing)

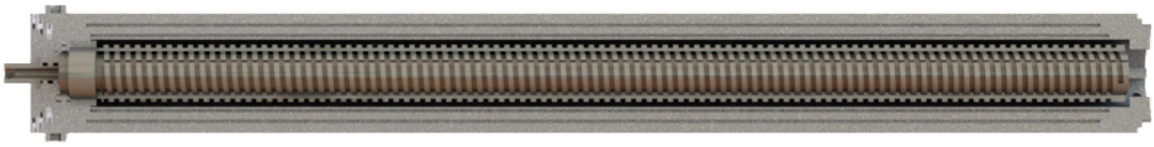
$\frac{F_a}{C_o}$ = 0.048 => e = 0.24 => x= 0.53 and y = 1.85 => Fe = 449.25

=> L10 = 10864.38 million revolution (Rating Life)

Selected Bearing - 6004 (Deep groove ball bearing, SKF catalogue)

Why choose screw threaded mechanism over other alternatives ?

- The screw-thread mechanism offers several advantages when utilized as a structural component for both the shaft and the extending mechanism.
- One notable benefit is its ability to eliminate the need for a separate extending mechanism, streamlining the overall design. Additionally, the self-locking nature of threads enables the rotation of the impeller, further enhancing the simplicity of the mechanism.
- By harnessing the full potential of a mechanical system and actuation method, the reliance on electrical components within the slurry is eliminated. This has several significant advantages, including the avoidance of water and spark-proofing requirements.
- As a result, the complexity of the system is reduced, and both manufacturability and serviceability are improved.



8. REFERENCES

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